

Topic 10 — Human-in-the-Loop Approval Workflow

Agent Orchestration, Alerting, and Human-in-the-Loop — Implementation & Validation Documentation

Project	HITL Content Generation
Platform	n8n
AI Provider	Google Gemini (Gemini 2.5 Flash)
Human Review Channel	Slack — #new-channel
Approval Mechanism	Slack Send and Wait for Approval
Error Handler	Topic 10 - Global Error Handler
Status	Tested successfully; main HITL workflow restored inactive
Documentation Date	13 September 2026

1. Assessment Overview

This project implements a production-oriented Human-in-the-Loop (HITL) approval workflow in n8n. AI-generated marketing content is sent to a human reviewer in Slack. The workflow pauses until the reviewer chooses approval or rejection, then routes the request to the correct branch. A centralized Error Trigger workflow provides operational alerting through Slack.

Assessment objectives addressed

- AI content generation with safety-oriented system instructions.
- LLM output controls and request timeout configuration.
- Human review using Slack Send and Wait for Approval.
- Workflow pause/resume using the approval interaction.
- Deterministic approval/rejection routing.
- Rejection handling that prevents publication.
- Centralized Error Trigger and Slack error notification.
- Live validation using real Slack interactions and a production-mode error test.

2. Solution Architecture

The final solution is split into a main HITL workflow and a separate global error-handler workflow.

Start → 1. Set Content Request → 2. AI Writes Draft → 3. Ask Human to Approve → 4. Approved?
TRUE → 5. Publish Approved Content
FALSE → 6. Prepare Rejection → 7. Send Rejection Notice

Error path: Production workflow failure → Error Trigger → Format Error Details → Slack Error Notification

3. Main Workflow Components

Node	Purpose
Start	Manual trigger used for controlled assessment testing.
1. Set Content Request	Creates request_id, request_type, topic, and user_request.
2. AI Writes Draft	AI Agent generates concise, safe, ready-to-publish content.
Google Gemini Chat Model	Provides Gemini generation for the AI Agent.
3. Ask Human to Approve	Posts the draft to Slack and pauses execution until a reviewer responds.
4. Approved?	Checks the boolean approval result and selects the correct branch.
5. Publish Approved Content	Publishes the approved draft to Slack.
6. Prepare Rejection	Creates REJECTED status and captures reviewer comments.
7. Send Rejection Notice	Notifies Slack that the draft was rejected and will not be published.

4. AI Safety and Generation Controls

The AI Agent is configured with a safety-focused system instruction. The prompt directs the model to produce professional content, avoid unsafe or non-compliant material, avoid inventing statistics/prices/testimonials/partnerships, avoid personal data and guarantees, and return only the final content.

Control	Configured value / behavior
AI provider	Google Gemini
Model	Gemini 2.5 Flash
Maximum output tokens	400
Request timeout	15,000 ms
Content limit	Maximum 350 words in the AI system instruction
Slack preview safety	Approval preview truncated to 2,800 characters

5. Human-in-the-Loop Approval

The Slack approval node uses the Send and Wait for Approval operation with double approval options. The generated draft is displayed to the reviewer. The workflow remains paused until the reviewer selects Approve & Publish or Reject and may provide an optional comment.

Approval path

Slack review → approved = true → Approved? TRUE branch → Publish Approved Content

The live approval test was completed successfully. The workflow paused at the approval step, a real Slack approval was submitted, the approval value evaluated to true, and the final content was published to Slack.

6. Rejection Routing

The rejection path is explicitly separated from the publishing path. When the reviewer rejects the draft, the IF node evaluates the approval value as false. The workflow then prepares a REJECTED status and sends a rejection notification. The publish node is not executed.

Slack review → approved = false → Approved? FALSE → Prepare Rejection → Send Rejection Notice

The live rejection test was completed successfully. Slack displayed the rejection notice with REJECTED status and stated that the draft would not be published.

7. Centralized Error Handling

A separate workflow named Topic 10 - Global Error Handler is assigned as the main workflow's error workflow. It contains three nodes:

Error Trigger → Format Error Details → Slack Error Notification

Node	Function
Error Trigger	Receives production execution failure information.
Format Error Details	Extracts workflow name, execution ID, error message, and timestamp.
Slack Error Notification	Posts the formatted operational alert to #new-channel.

8. Live Error-Handler Validation

A controlled production-mode test was performed using a temporary isolated webhook test path. The main workflow failed intentionally at a temporary force-error step. The production failure triggered the assigned global error workflow, which completed successfully and posted a real Slack alert.

Confirmed test chain: production execution 203 → error workflow execution 204 → Slack Error Notification.

The Slack alert contained the workflow name, execution ID, controlled error message, and timestamp. After validation, the temporary webhook and force-error step were removed and the main HITL workflow was restored to its working state.

9. Validation & Test Evidence

Test	Expected result	Result
AI draft generation	Gemini generates safe draft	PASS
HITL pause	Workflow waits for human response	PASS
Approval routing	TRUE → Publish Approved Content	PASS
Approval publication	Approved content reaches Slack	PASS
Rejection routing	FALSE → rejection branch	PASS
No publication after rejection	Publish node not executed	PASS
Global error workflow	Production error triggers handler	PASS
Slack error notification	Error details posted to Slack	PASS

10. Evidence Files Recommended for Submission

The following evidence should accompany this documentation in the public repository:

- approval-success.png — Slack approval and n8n TRUE branch evidence.
- rejection-success.png — Slack rejection and n8n FALSE branch evidence.
- error-handler-success.png — real production error alert in Slack.
- approval-live-test.mp4 — captured live approval test recording.
- rejection-live-test.mp4 — captured live rejection test recording.

11. Final Submission Structure

Recommended public repository structure:

```
topic-10-hitl-approval-workflow/ ■■■■ README.md ■■■■ workflows/ ■ ■■■■ HITL Content
Generation.json ■ ■■■■ Topic 10 - Global Error Handler.json ■■■■ docs/ ■ ■■■■ Topic 10 HITL
Approval Workflow Documentation.pdf ■■■■ screenshots/ ■ ■■■■ approval-success.png ■ ■■■■
rejection-success.png ■ ■■■■ error-handler-success.png ■■■■ test-evidence/ ■■■■
approval-live-test.mp4 ■■■■ rejection-live-test.mp4
```

12. Final Assessment Checklist

- AI Agent added and configured — COMPLETE
- Gemini Chat Model connected — COMPLETE
- Timeout 15,000 ms — VERIFIED
- Max output tokens 400 — VERIFIED
- Slack review notification — COMPLETE
- Human-in-the-Loop pause/resume — VERIFIED LIVE
- Approval routing — VERIFIED LIVE
- Rejection routing — VERIFIED LIVE
- Rejected content prevented from publishing — VERIFIED

- Global Error Handler — COMPLETE
- Error Trigger — VERIFIED with production failure
- Slack error notification — VERIFIED LIVE
- Workflow JSON exports — READY
- Documentation — THIS PDF
- Screenshots and test evidence — READY TO UPLOAD
- Loom/YouTube demo — TO BE RECORDED FROM EXISTING EVIDENCE

Note: This document describes the final tested implementation and is intended to accompany the exported n8n workflow JSON files and captured test evidence.